Mohd Atif

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Summary

Experienced Data Scientist with 1 year of work experience in creating ML pipelines, time-series models, and LLM applications using Python. Accomplishments include a 20-35% increase in forecast accuracy, 15% increase in gross margin, and 95% classification accuracy on a deployed system. Experienced in all aspects of the data science process - from data engineering to model deployment. Familiar with programmatic integration of an LLM API, development of an agent, and prompt engineering.

Professional Experience

Data Scientist — *Arabania Indo-Arabic Restaurant*

June 2024 - May 2025, Lucknow

- Built demand forecasting models in Python using statsmodels and scikit-learn across 100K+ inventory records, helping reduce food waste by 20% and procurement costs by 12% through more accurate daily inventory planning.
- Ran end-to-end A/B pricing experiments using hypothesis testing and statistical inference, identifying a pricing strategy that improved gross margins by 15% within 2 months, while also building SKU performance models that improved operational efficiency by 12%.
- Built automated SQL analytics pipelines using CTEs and window functions to uncover demand and seasonality patterns across 100K+ records, reducing manual reporting effort by 40% and making recurring analysis more efficient.

Projects

European Power Market Forecasting System — GitHub

Python, XGBoost, ENTSO-E API, LLM Integration, Prompt Engineering, PnL Simulation

- Built an end-to-end electricity price forecasting pipeline on real ENTSO-E European market data, engineering 12+ features (lag-1/24/168h, rolling volatility, calendar effects) and training an XGBoost model achieving 20-35% improvement over baseline, outperforming 3 alternative models evaluated during development.
- Programmatically integrated an LLM via API with structured prompt engineering to build a custom agent auto-generating trader briefs from model outputs, eliminating 3+ hours of manual report writing per week across 5 recurring report types.
- Designed an 8-point automated QA pipeline (schema validation, anomaly detection, correlation checks, freshness monitoring) reducing data quality incidents to zero, and implemented PnL simulation and backtesting framework across 12+ months of historical market data.

Customer Churn Prediction and Retention Engine — Live App

Python, FastAPI, Scikit-learn, LLM API Integration, Streamlit

- Trained and deployed a Logistic Regression classifier achieving 95% accuracy on behavioral features (usage, support tickets, tenure), serving real-time inference to 100+ customers via a production FastAPI backend with batch CSV processing.
- Built a custom LLM agent via API that interprets churn scores and feature importances to generate multi-step retention recommendations; implemented explainability layer surfacing top 3 churn drivers per prediction.
- Designed What-If simulation framework enabling stakeholders to evaluate intervention ROI independently, reducing ad-hoc analyst requests by providing self-service scenario analysis.

Meteograph — GitHub

Python, FastAPI, JavaScript, OpenWeatherMap, TomTom, OpenStreetMap

- Built a full-stack weather and smart-city dashboard using a modular FastAPI backend and static HTML/CSS/JavaScript frontend, integrating weather, air quality, traffic, geocoding, alerts, and forecast data into a unified location-based application.
- Architected server-side API services with centralized configuration, response models, short-lived TTL caching, and automatic mock-data fallback, keeping external API keys off the client while allowing the application to run end-to-end without API credentials.
- Developed interactive weather visualizations, animated weather scenes, pollutant breakdowns, traffic metrics, trend charts, map-based location selection, and 24-hour/7-day forecasts, with live data support through OpenWeatherMap and TomTom.

Technical Skills

- **ML and Statistics:** XGBoost, scikit-learn, Logistic Regression, time-series forecasting, feature engineering, cross-validation, A/B testing, anomaly detection, backtesting, PnL simulation, statistical inference
- **LLMs and Agents:** Programmatic LLM API integration (OpenAI, Anthropic), LangChain, RAG pipelines, custom agent development, prompt engineering, agentic workflows, structured output validation
- **Programming and Data:** Python (Pandas, NumPy, FastAPI, Streamlit, statsmodels), SQL (CTEs, window functions, query optimisation), R, Git, REST APIs, automated data pipelines
- **Visualisation and Tools:** Power BI, Tableau, Matplotlib, Seaborn, Jupyter, data documentation

Certifications

- Google Data Analytics Professional Certificate (Foundations, Data Exploration, Data-Driven Decisions)

Education

Indian Institute of Technology, Madras — *2022-2025*

B.S. in Data Science and Applications